Module-3
PUBLICATION ETHICS AND MISCONDUCTS

Ethics: definition; Ethics with respect to science and research - intellectual honesty and research integrity; Scientific misconducts: Falsification, Fabrication and Plagiarism (FFP); Redundant publications: duplicate and overlapping publications, salami slicing; Selective reporting and misrepresentation of data; Publication ethics: introduction and importance; Standards setting initiatives and guidelines: COPE, WAME; Conflicts of interest. Publication misconduct: definition, concept, unethical behaviour; Violation of publication ethics, authorship and contributor ship; Identification, complaints and appeals; Predatory publisher and journals.

Ethics

- Everybody in their everyday life attempts to behave from certain perspectives laid out moral standards. This **philosophical idea** has various applications in an **individual's reality**.
- Morals is concerned with the meaning of good and bad. It clarifies ways of thinking that teach us the proper behaviour in a given circumstance, which has forever involved dispute between savants.
- Each thinker has characterized it as indicated by their own emotional understanding. While epistemology is worried about what we should accept and how we should reason.
- **Ethics is worried about what we should do, how we should live, and how we should coordinate our networks.** Unfortunately, it shocks numerous new truth seekers that can reason about things like this. Strictly propelled sees about ethical quality frequently take right and wrong to be only a question of what is instructed by a heavenly being.
- Moral Relativism, maybe the most prominent attitude among individuals who have dismissed confidence, essentially substitutes the orders of society for the orders of God.

- **Considering ethical quality** as far as whose orders are definitive rules out level headed investigation into how we should live, how we should treat others, or how we should structure our networks.
- Reasoning, then again, treats in a serious way the chance of objective request into these issues.
- The long furthermore, troublesome history of science ought to provide us with some modest acknowledgment of how troublesome and disappointing cautious request and examination can be.
- The way of thinking of science, for example, is worried about magical issues about what science is, yet in addition with epistemological inquiries concerning how we can know logical insights.
- The way of thinking of affection is also worried about magical inquiries concerning what love is. However, it additionally worried about inquiries regarding the worth of adoration that are more moral in character.

Ethics Defined

- A discipline dealing with what is proper course of action for man (Aristotle, *cit in* Mckeon,1941)
- A branch of philosophy that looks at what is good and what is bad
- A system of obligation that we have towards others
- Also known as moral philosophy, involves, systematising, *defending*, and *recommending* concepts of right and wrong behaviour *(www.iep.utm.edu/ethics)*
- *A study of principles guiding the good of the individual within the context of social interactions and the community*

Type of Ethics - Defined

ETHICS AS A THEORETICAL ENTERPRISE

Meta Ethics: "*is a branch of analytic philosophy that explores the status, foundations, and scope of moral values, properties, and words*" *Source:* https://www.princeton.edu/~achaney/tmve/wiki100k/docs/Meta-ethics.html

Normative Ethics: addresses the question of 'What ought to be done?' Normative Ethical theorists (Socrates, Kant, Stuart, Bentham) seek to provide action guides and codes.

Applied Ethics: domain specific ethics (Business, biomedical; APA, engineering, etc.)

Ethics Evolution

Aristotle (384-322 BC) – proposed a theory of virtue

Socrates (469-399)

Siddhartha Gautama (563-480 BC)

Kant (1724-1804)

rightness of an action is determined by the character of the principle that a person chooses to act upon

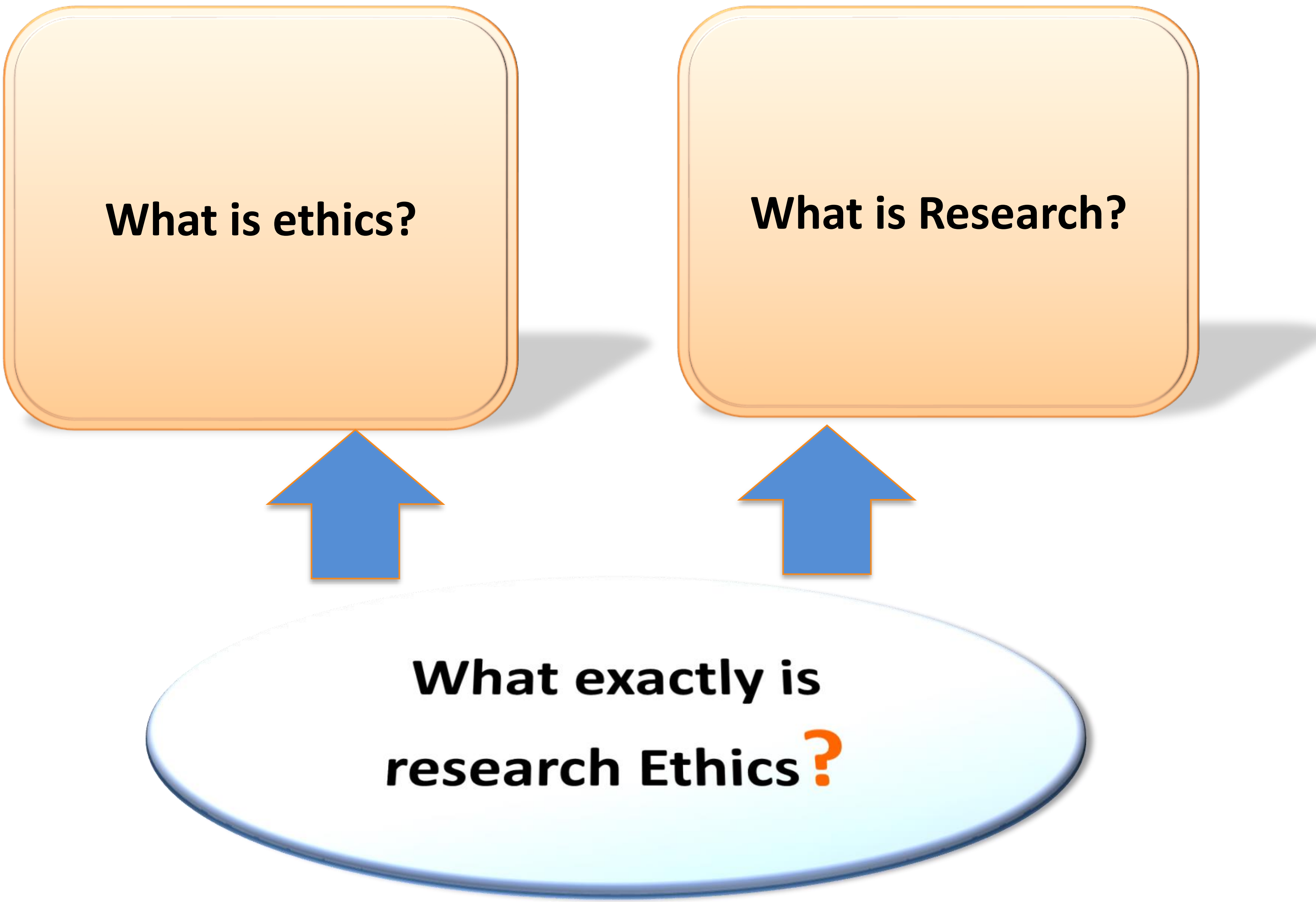
Ethics Evolution

- Bentham (1781) – Principle of utility – diapproves or approves of every action according to tendency to augment or diminish happiness
- Mahatma Ghandi (1869 – 1948)
- Martin Luther King

  Nelson Mandela (1918-2013)
- Fanon (1925-1961) African philosophy, humaneness

What is Research Ethics?

- ***Ethics*** are the **set of rules** that govern our expectations of our own and others' behavior.
- ***Research ethics*** are the **set of ethical guidelines** that guides us on how ***scientific research* should be *conducted* and *disseminated***.
- Research ethics govern the standards of conduct for scientific researchers .
- It is the guideline for responsibly conducting the research.
- Research that implicates human subjects or contributors rears distinctive and multifaceted ethical, legitimate, communal and administrative concerns.
- Research ethics is unambiguously concerned in the examination of ethical issues that are upraised when individuals are involved as participants in the study.
- Research ethics committee/Institutional Review Board (IRB) reviews whether the research is ethical enough or not to protect the rights, dignity and welfare of the respondents.


What is ethics?
What is Research?
What exactly is research Ethics?

Research Ethics

"Greek ***ethos*** 'character' is the systematic study of value concepts—good, bad, right, wrong and the general principles that justify applying these concepts".

Joan E. Sieber

Planning Ethically Responsible Research, p. 3

Research Ethics

"Ethics is the disciplined study or morality….and morality asks the question…what should one's behavior be".

Jeff Cooper

Albany Medical Center, Ethical Decision Making, 2001, p. 1

Research Ethics

Two types of ethical decision-making

1. Deductive or principle based reasoning
2. Inductive or case based reasoning

Jeff Cooper

Albany Medical Center, Ethical Decision Making, 2001, p. 1

Research Ethics

1. Deductive or principle based reasoning
   – Start with an ethical theory
   – Continue with a specific principle
   – Develop rules
   – Make judgments

Jeff Cooper

Albany Medical Center, Ethical Decision Making, 2001, p. 1

Research Ethics

1. Deductive Reasoning

Ethical Theory → Principle → Rules → Judgment

Peace → Violence → War

Jeff Cooper

Albany Medical Center, Ethical Decision Making, 2001, p. 2

Research Ethics

2. Inductive or Case Based Reasoning

- Decisions we have made – precedent
- Look back at those decisions and combine them in order to make a judgment
- Judgments reflect back on rules
- Rules reflect on our principles
- Principles reflect back to the ethical theory

Jeff Cooper

Albany Medical Center, Ethical Decision Making, 2001, p. 2

Research Ethics

2. Inductive or Case Based Reasoning

- Decisions we have made – avoid war and move to Canada (U.S. declares war on Canada)
- Judgment – defend yourself
- Rule – join Army (protect children)
- Principles – family important
- Ethical theory

Jeff Cooper
Albany Medical Center, Ethical Decision making, 2001, p. 2

Research Ethics

Deductive reasoning → Case Based Reasoning

No WAR! Conflict Fight WAR!

Conflict Between Decisions

- When there is an argument
- Go back to the original principles – ask yourself “What were my original principles?”
- Original principles are in conflict or “incoherent”

Jeff Cooper

Albany Medical Center, Ethical Decision Making, 2001, p. 3

Research Ethics

Conflict Between Decisions

- There will be conflict
- You will use both types of ethical decision-making to make decisions
- When conflict arises…go back to the original principles and try to create coherence by dealing with the specific principles

Jeff Cooper

Albany Medical Center, Ethical Decision Making 2001, p. 3

Key Features of Scientific Research

Generalisability of findings: the extent to which the sample used in the research project reflects the broader population of interest

Scientific Rigour (truth is accepted if there is sufficient evidence to support claims made through the research process. Such claims have to withstand the scrutiny of repeated testing)

Univesality and objectivity (explicit rules and systematic procedures)
- Research should be designed in a manner that allows any competent researcher to conduct a similar study and generate the same findings

Originality of research work: original ideas backed with appropriate evidence in a clear, logical and convincing argument that illustrates critical and analytical thinking.

Research Ethics are:

1. A code of guidelines on how to conduct scientific research in a morally acceptable way.

2. Principles and standards that help researchers to uphold the value and standards of knowledge construction.

Ethical Considerations in the Research Process

Ethical considerations come into play at six stages of research

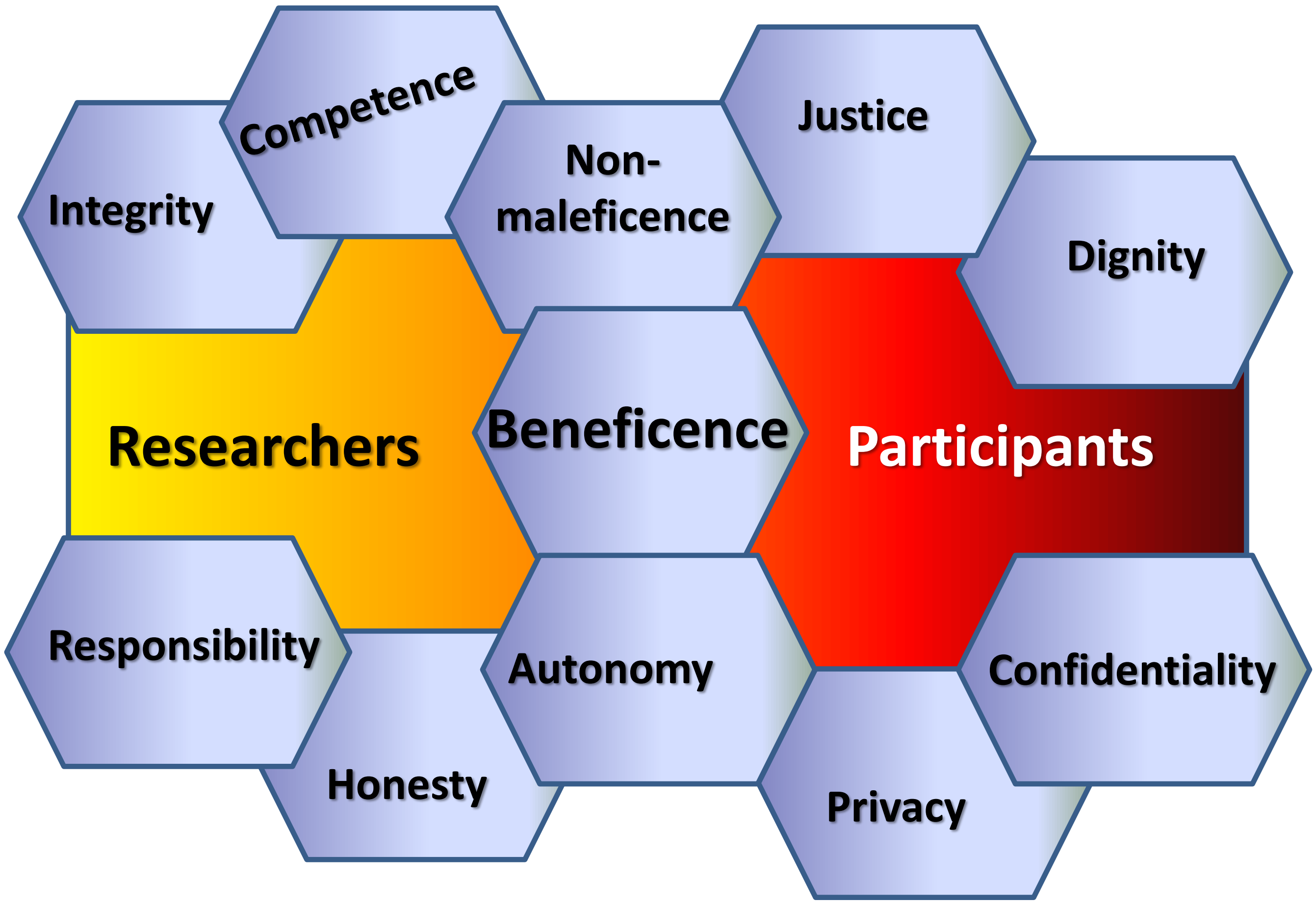
1. **Conceptualisation and design of the study (scientific merit, identify risks and ways to mitigate the risks)**
2. **When participants are recruited (the process of informed consent, right to privacy)**
3. **During the intervention or measurement procedure to which participants are subjected (management of risk)**
4. **In the release of results obtained**
5. **(protection of confidentiality and anonimity)**
6. **After the release of results (ensure that participants and communities involved in the research benefit)**

Objectives of Research Ethics

- **Three comprehensive objectives of Research Ethics are –**

1. To guard/protect human participants, their dignity, rights and welfare .
2. To make sure that research is directed in a manner that assists welfares of persons, groups and/or civilization as a whole.
3. To inspect particular research events and schemes for their ethical reliability, considering issues such as the controlling risk, protection of privacy and the progression of informed consent.

Principles of Research Ethics



Principles of Research Ethics

The general principles of research ethics are:

- **Honesty**
  - Being honest with the beneficiaries and respondents.
  - Being honest about the findings and methodology of the research.
  - Being honest with other direct and indirect stakeholders.
- **Integrity**
  - Ensuring honesty and sincerity.
  - Fulfilling agreements and promises.
  - Do not create false expectations or make false promises.
- **Objectivity**
  - Avoiding bias in experimental design, data analysis, data interpretation, peer review, and other aspects of research.
- **Informed consent**
  - Informed consent means that a person knowingly, voluntarily and intelligently gives consent to participate in a research.
  - Informed consent is related to the autonomous right of the individual to participate in the research.
  - Informing the participant about the research objective, their role, benefits/harms (if any) etc.

Principles of Research Ethics

- The general principles of research ethics are:
- **Respect for person/respondent**
  - Autonomy, which requires that those who are capable of deliberation about their personal goals should be treated with respect for their capacity for self-determination; and
  - Protection of persons with impaired or diminished autonomy, which requires that those who are dependent or vulnerable be afforded security against harm or abuse.
- **Beneficence**
  - Maximize the benefits of the participants. Ethical obligation to maximize possible benefits and to minimize possible harms to the respondents.
- **Non-maleficence/ Protecting the subjects (human)**
  - Do no harm. Minimize harm/s or risks to the human. Ensure privacy, autonomy and dignity.
- **Responsible publication**
  - Responsibly publishing to promote and uptake research or knowledge. No duplicate publication.
- **Protecting anonymity**
  - It means keeping the participant anonymous. It involves not revealing the name, caste or any other information about the participants that may reveal his/her identity.

Principles of Research Ethics

- The general principles of research ethics are:
- **Confidentiality**
  - Protecting confidential information, personnel records. It includes information such as:
  - Introduction and objective of the research
  - Purpose of the discussion
  - Procedure of the research
  - Anticipated advantages, benefits/harm from the research (if any)
  - Use of research
  - Their role in research
  - Right to refuse or withdraw
  - Methods which will be used to protect anonymity and confidentiality of the participant
  - Freedom to not answer any question/withdraw from the research
  - Who to contact if the participant needs additional information about the research.

Principles of Research Ethics

- The general principles of research ethics are:
- **Non-discrimination**
  - Avoid discrimination on the basis of age, sex, race, ethnicity or other factors that are violation of human rights and are not related to the study.
- **Openness**
  - Be open to sharing results, data and other resources. Also accept encouraging comments and constructive feedback.
- **Carefulness and respect for intellectual property**
  - Be careful about the possible error and biases.
  - Give credit to the intellectual property of others. Always paraphrase while referring to others article, writing. Never plagiarize.
- **Justice**
  - The obligation to distribute benefits and burdens fairly, to treat equals equally, and to give reasons for differential treatment based on widely accepted criteria for just ways to distribute benefits and burdens.

Broad Categorization of Research Ethics

1. MINIMIZING THE RISK OF HARM
2. OBTAINING INFORMED CONSENT
3. PROTECTING ANONYMITY AND CONFIDENTIALITY
4. AVOIDING MISLEADING PRACTICES
5. PROVIDING THE RIGHT TO WITHDRAW

Broad Categorization of Research Ethics

1. MINIMIZING THE RISK OF HARM

- It is necessary to minimize any sort of harm to the participants. There are a number of forms of harm that participants can be exposed to. They are:
  - Bodily harm to contributors.
  - Psychological agony and embarrassment.
  - Social drawback.
  - Violation of participant's confidentiality and privacy.
- In order to minimize the risk of harm, the researcher/data collector should:
  - Obtain **informed consent** from participants.
  - Protecting anonymity and confidentiality of participants.
  - Avoiding **misleading practices** when planning research.
  - Providing participants with the **right to withdraw**.

Broad Categorization of Research Ethics

2. OBTAINING INFORMED CONSENT

- One of the fundamentals of research ethics is the notion of **informed consent**.
- Informed consent means that a **person knowingly, voluntarily and intelligently gives consent to participate in a research.**

Informed consent means that the participants should be well-informed about the:

- Introduction and objective of the research
- Purpose of the discussion
- Anticipated advantages, benefits/harm from the research (if any)
- Use of research
- Their role in research
- Methods which will be used to protect anonymity and confidentiality of the participant
- Freedom to not answer any question/withdraw from the research
- Who to contact if the participant need additional information about the research

Broad Categorization of Research Ethics

<u>3. PROTECTING ANONYMITY AND CONFIDENTIALITY</u>

Protecting the **anonymity** and **confidentiality** of research participants is an additionally applied constituent of research ethics.

- **<u>Protecting anonymity:</u>** It means keeping the participant anonymous. It involves not revealing the name, caste or any other information about the participants that may reveal his/her identity.
- **<u>Maintaining confidentiality:</u>** It refers to ensuring that the information given by the participant are confidential and not shared with anyone, except the research team. It is also about keeping the information secretly from other people.

<u>4. AVOIDING MISLEADING PRACTICES</u>

- The researcher should avoid all the deceptive and misleading practices that might misinform the respondent.
- It includes avoiding all the activities like communicating wrong messages, giving false assurance, giving false information etc.

Broad Categorization of Research Ethics

5. PROVIDING THE RIGHT TO WITHDRAW

- Participants have to have the right to withdraw at any point of the research.
- When any respondent decides on to withdraw from the research, they should not be **stressed** or **forced** in any manner to try to discontinue them from withdrawing.

Apart from the above-mentioned ethics, other ethical aspects things that must be considered while doing research are:

Protection of vulnerable groups of people:

- Vulnerability is one distinctive feature of people incapable to protect their moralities and wellbeing. Vulnerable groups comprise captive populations (detainees, established, students, etc.), mentally ill persons, and aged people, children, critically ill or dying, poor, with learning incapacities, sedated or insensible.
- Their participation in research can be endorsed to their incapability to give an informed consent and to the need for their further safety and sensitivity from the research/researcher as they are in a greater risk of being betrayed, exposed or forced to participate.

Skills of the researcher:

- Researchers should have the basic skills and familiarity for the specific study to be carried out and be conscious of the bounds of personal competence in research.
- Any lack of knowledge in the area under research must be clearly specified.
- Inexperienced researchers should work under qualified supervision that has to be revised by an ethics commission.

Advantages of Research Ethics

- **Research ethics promote the aims of research.**
- **It increases trust among the researcher and the respondent.**
- It is important to adhere to ethical principles in order to protect the dignity, rights and welfare of research participants.
- **Researchers can be held accountable and answerable for their actions.**
- **Ethics promote social and moral values.**
- **Promotes the** ambitions of research, such as understanding, veracity, and dodging of error.
- Ethical standards uphold the **values that are vital to cooperative work**, such as belief, answerability, mutual respect, and impartiality.
- Ethical norms in research also aid to construct **public upkeep** for research. People are more likely to trust a research project if they can trust the worth and reliability of research.

Limitations of Research Ethics

- Possibilities to physical integrity, containing those linked with experimental drugs and dealings and with other involvements that will be used in the study (e.g. measures used to observe research participants, such as blood sampling, X-rays or lumbar punctures).
- **Psychological risks:** for example, a questionnaire may perhaps signify a risk if it fears traumatic events or happenings that are especially traumatic.
- **Social, legal and economic risks**: for example, **if** personal information collected during a study is unintentionally released, participants might face a threat of judgment and stigmatization.
- Certain tribal or inhabitant groups may possibly suffer from discrimination or stigmatization, burdens because of research, typically if associates of those groups are recognized as having a greater-than-usual risk of devouring a specific disease.
- The research may perhaps have an influence on the prevailing health system: for example, human and financial capitals dedicated to research may distract attention from other demanding health care necessities in the community.

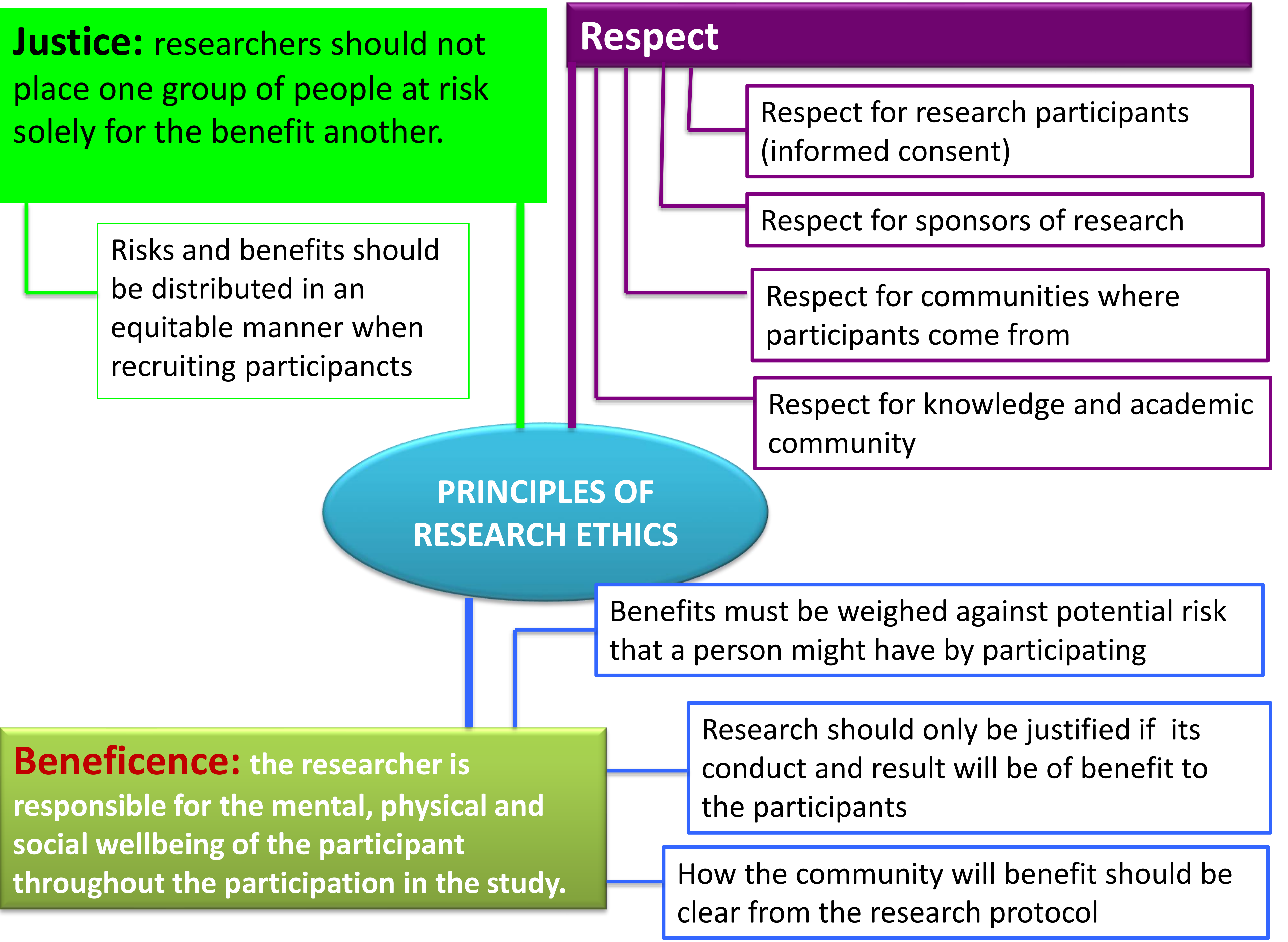
Ensure Ethics at Different Steps of Research

The following process helps to ensure ethics at different steps of research:

- Collect the facts and talk over intellectual belongings openly
- Outline the ethical matters
- Detect the affected parties (stakeholders)
- Ascertain the forfeits – Check different forces
- Recognize the responsibilities (principles, rights, justice)
- Contemplate your personality and truthfulness
- Deliberate innovatively about possible actions
- Respect privacy and confidentiality
- Resolve on the appropriate ethical action and be willing to deal with divergent point of view.

FUNDAMENTALLY Research Ethics are:

- A way of conducting the research enterprise such that the three fundamental principles of research (respect, beneficence and justice) are upheld.
- Ethical research must conform with the national and international accords and prescripts.


Justice: researchers should not place one group of people at risk solely for the benefit another.
Risks and benefits should be distributed in an equitable manner when recruiting participancts
Respect
Respect for research participants (informed consent)
Respect for sponsors of research
Respect for communities where participants come from
Respect for knowledge and academic community
PRINCIPLES OF RESEARCH ETHICS
Benefits must be weighed against potential risk that a person might have by participating
Research should only be justified if its conduct and result will be of benefit to the participants
How the community will benefit should be clear from the research protocol
Beneficence: the researcher is responsible for the mental, physical and social wellbeing of the participant throughout the participation in the study.

Concerned with Research Ethics: Reasons

1. Professional Responsibility
2. To avoid reputational damage
3. Research can be harmful to:
   – Participants
   – To researchers
   – To institutions
   – To research communities

4. To avoid litigation
   – In a scenario where a proposal is classified as Ethics Category 1 (exempt from Ethics and Biosafety Research Committee Review) liability and responsibility arising from decisions based on ethics are shouldered by the FRC and its members.

Six Fundamental Ethical Principles for Scientific Research

Attribution of credit

Scientists should not plagiarise the work of other scientists. They should give credit where credit is due but not where it is not due.

Six Fundamental Ethical Principles for Scientific Research

Public responsibility

Scientists should report research in the public media when the research has an important and direct bearing on human happiness and when the research has been sufficiently validated by scientific peers.

Ethical considerations for scientific conduct

- All behaviors involved in the research process, such as developing a theory, collecting data, and testing hypotheses, are subject to ethical considerations, codified and uncodified**, particularly ethics related to empirical data collection and human subjects**.

Ethical considerations for scientific conduct

- Research involving human subjects in institutions that receive federal research funding must receive **ethical clearance** by an **independent review board**.
- IRB must approve any research with human subjects *before* it is initiated.

Ethical clearance considerations

- An IRB evaluates
1. the extent to which participation in a study
    - is voluntary,
    - does not exert physical or psychological stress, and
    - not cause other kinds of damage to participants
2. whether participants must give consent regarding
    - how their data will be used
    - how their data will be reported
    - how the data will be protected in terms of anonymity or confidentiality
3. whether participants have the right to withdraw from participation at any time.
4. how data is stored and analysed
    - Involves ownership, storage and backup, privacy, confidentiality, access, and reuse.

Another way of looking at research ethics is by looking at unethical research conduct

- Deception (issues of full disclosure)
    - Withholding information about the aim of the study
    - Misleading participants about the risks inherent in participating in the study
- Plagiarism
- Conducting research that does not have a scientific base (ill-formed problem statement)
- Lack of objectivity and integrity in the design and conduct of research
    - Not identifying the methodological constraints of the study that determine the validity of the findings
    - Misinterpretation of results
    - Not providing details of theories and methods that might be relevant in the interpretation of research findings
- Fabrication or falsification of data
- Not following the appropriate ascription of authorship to a publication

Another way of looking at research ethics is by looking at unethical research conduct

- Not respecting the right to privacy
- Not respecting the right to anonymity and confidentiality
- Not respecting rights of vulnerable groups
  - Children
  - Mentally handicapped individuals
  - The aged
  - Prisoners
  - Illiterate
  - Those with low social status
- Not having due consideration for the environment

Publication Ethics:
Ethical considerations for scientific writing

- A subset of ethical issues in scientific conduct that relates only to the reporting of research
- Very important part of scientific ethics because it is typically only through reported research that an ethical issue is revealed
  - we typically cannot learn about data fabrication or amendment until those data are disclosed.
  - We cannot identify a lack of attribution of credit until an unnamed contributor sees it in writing

Scientific misconducts:
Falsification, Fabrication and Plagiarism (FFP)

Scientific misconduct refers to unethical practices in research that compromise the integrity of scientific work. The three primary forms of scientific misconduct are **Falsification**, **Fabrication**, and **Plagiarism (FFP)**. These actions violate the fundamental principles of research ethics and can undermine public trust in science.

Falsification

Falsification involves manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record.

•**Examples**:

- Adjusting data to fit a desired outcome.
- Manipulating images (e.g., changing the contrast in gel electrophoresis images to make results seem more significant).
- Omitting negative or contradicting data from a study to present a biased result.

Falsification can lead to misleading conclusions, which may cause harm if applied in fields like medicine or environmental science.

Fabrication

Fabrication refers to making up data or results and recording or reporting them as if they were real.

•**Examples**:

- Creating fake data for experiments that were never conducted.
- Reporting findings from nonexistent surveys or clinical trials.

Fabrication leads to a complete distortion of scientific knowledge, potentially influencing policy, medical treatments, or future research based on non-existent data.

Plagiarism

Plagiarism involves taking someone else’s ideas, processes, results, or words without giving appropriate credit. This can include copying text, figures, or entire sections of research from another source.

•**Examples**:

- Copying text from another researcher’s paper without citation.
- Using someone else’s research idea or methodology without permission or acknowledgment.
- Submitting the same research paper to different journals without disclosure (self-plagiarism).

Plagiarism disrespects the intellectual property of other researchers and compromises the originality of the research process.

Consequences of FFP:

•**Damage to Reputation**: Researchers found guilty of FFP often face public scrutiny, lose their credibility, and may be barred from publishing in reputable journals.
•**Retraction of Papers**: Journals may retract papers that are found to involve misconduct, impacting the entire research community relying on those results.
•**Legal or Career Repercussions**: In some cases, research misconduct can lead to job loss, legal action, or the revocation of research funding.

Prevention:

•**Rigorous Peer Review**: Peer review processes aim to detect misconduct before publication.
•**Ethical Training**: Many institutions require researchers to undergo training in responsible research conduct.
•**Transparency in Research**: Open data, materials, and methods allow for better scrutiny of research findings.

Redundant publications: duplicate and overlapping publications, salami slicing

Redundant publications refer to the practice of publishing the same or substantially similar research in multiple places without proper disclosure. This unethical practice can inflate a researcher's publication count while wasting the time of peer reviewers and distorting the academic record. There are three main types of redundant publications.

Duplicate Publications

Duplicate publication occurs when the same work (or nearly identical work) is published more than once without notifying the readers, editors, or reviewers.

•**Examples**:

- Publishing the same research in two different journals without indicating that the paper has been previously published.
- Submitting the same manuscript to multiple journals simultaneously (also known as "simultaneous submission").

Duplicate publications can mislead the scientific community by giving the impression that a particular finding has been independently verified or reproduced, when in fact it is the same study being counted multiple times. It can also lead to wasted resources by journals and reviewers.

Overlapping Publications

Overlapping publication occurs when a manuscript shares significant content with another paper but is presented as a new study, without proper citation or disclosure of the overlap.

•**Examples**:

- Using the same data set to draw conclusions for two different papers without clarifying the overlap.
- Reusing substantial portions of the text from a previous publication without proper acknowledgment (self-plagiarism).

Overlapping publications distort the perception of novelty and can cause confusion in the literature, as researchers may believe they are citing new findings when in reality they are referencing the same work repeatedly.

Salami Slicing (Segmented Publication)

Salami slicing refers to the practice of dividing a large study into several smaller papers that each report part of the findings, even though they would be better presented as a single, comprehensive study.

•**Examples**:

- A researcher conducting a large survey but publishing individual papers on separate variables from the same data set instead of reporting the full findings together.
- Splitting the results of an experiment into multiple publications, where each paper contains incremental or partial data that, together, would form a more cohesive study.

Salami slicing can inflate a researcher’s publication record artificially and may dilute the scientific message. It can also lead to fragmented understanding, where readers may not fully grasp the overall implications of a study because the results are scattered across multiple papers.

Redundant Publications lead to …

•**Loss of Trust**: Journals and academic institutions take redundant publications seriously, and researchers found guilty may suffer reputational damage or face retraction of their papers.

•**Waste of Resources**: Redundant publications waste the time and efforts of editors, reviewers, and other researchers who rely on the integrity of the scientific record.

•**Distortion of the Literature**: By inflating the appearance of scientific progress, redundant publications can skew meta-analyses and literature reviews, leading to flawed conclusions.

How to avoid Redundant Publications

•**Clear Disclosure**: Researchers should disclose previous publications and any overlap when submitting manuscripts, ensuring that editors are aware of any related work.

•**Comprehensive Reporting**: Large studies should be published in their entirety, where appropriate, rather than broken into smaller, less meaningful parts.

•**Journal Policies**: Many journals now have strict guidelines against duplicate or overlapping publications and employ plagiarism detection software to catch such instances.

Selective reporting and misrepresentation of data

Selective reporting and misrepresentation of data are unethical practices in scientific research where findings are presented in a biased or misleading way. These practices can severely distort scientific knowledge, lead to erroneous conclusions, and undermine the credibility of research.

Selective Reporting

Selective reporting occurs when researchers choose to report only positive, significant, or desired outcomes, while ignoring or downplaying negative, non-significant, or conflicting results. This creates a biased representation of the research findings.

•**Examples**:

- Only publishing favorable results while withholding unfavorable data that doesn't support the hypothesis (also known as "cherry-picking").
- Reporting only statistically significant results without mentioning non-significant findings from the same study.
- Failing to disclose secondary or exploratory analyses that contradict the primary findings.

Distortion of the scientific record: Other researchers who rely on published results to build on existing knowledge may form inaccurate conclusions, leading to skewed literature reviews or meta-analyses.
Bias in clinical trials: In fields like medicine, selective reporting can have serious consequences, as it may give an exaggerated sense of a drug's efficacy or safety while underreporting adverse effects.

Misrepresentation of Data

Misrepresentation of data involves deliberately altering, misinterpreting, or presenting data in a misleading way to make the findings appear more favorable or significant than they actually are. This could involve manipulating graphs, altering figures, or presenting statistical results in a deceptive manner.

•**Examples**:

- **Data manipulation**: Adjusting data points or statistical analyses to achieve desired results (e.g., rounding numbers, removing outliers without justification).
- **Misleading graphs or visuals**: Creating visual representations of data that exaggerate differences, such as by manipulating the scales of graphs.
- **P-hacking**: Conducting multiple statistical tests until significant results are found, then reporting only those results, even if they are due to chance.

False conclusions: Misrepresentation can lead to the publication of false or misleading conclusions that may influence future research, policy decisions, or clinical guidelines.

Erosion of trust: When misrepresentation is uncovered, it not only damages the credibility of the researchers involved but also undermines public trust in scientific research as a whole.

Common Forms of Selective Reporting and Misrepresentation:

- **Reporting Bias**: This includes a range of biases like publication bias (where studies with positive results are more likely to be published), outcome reporting bias (reporting only significant outcomes), and analysis bias (only reporting certain analyses that yield desired results).
- **Incomplete Reporting**: In some cases, researchers may selectively report the methods and materials used in their study, which makes it difficult for others to replicate or verify the findings.
- **Overstating Findings**: Researchers may exaggerate the significance or implications of their results, often using terms like “breakthrough” or “significant” when the data does not fully support such conclusions.

Selective Reporting and Misrepresentation may cost …

- **Skewed Meta-analyses**: Systematic reviews and meta-analyses rely on published data to synthesize findings across studies. Selective reporting or misrepresentation can distort these reviews, leading to incorrect scientific conclusions.
- **Wasted Resources**: Misleading studies may cause researchers to follow unproductive lines of inquiry, wasting time, funding, and effort.
- **Ethical Violations**: In fields like clinical research, selectively reporting or misrepresenting data can put human lives at risk by hiding critical information about drug efficacy or side effects.

Hence, we need to do

- **Pre-registration of Studies**: Pre-registering studies and their hypotheses (especially in clinical trials) helps ensure transparency and reduces the likelihood of selective reporting.
- **Data Transparency**: Sharing raw data along with publications encourages scrutiny and validation of results by the broader research community.
- **Rigorous Peer Review**: Peer reviewers and journal editors should scrutinize methods and data presentation closely, asking for clarification when necessary.
- **Ethical Guidelines and Policies**: Institutions and funding agencies should enforce strict guidelines against selective reporting and misrepresentation, with consequences for those who violate them.

Plagiarism

- the wrongful appropriation, close imitation, or purloining and publication of another author’s language, thoughts, ideas, or expressions and their representation as one’s own work.
- the act of passing off someone else's work as your own, whether intentionally or unintentionally.
- The most common form of scientific misconduct.

Forms of plagiarism

- Intentional plagiarism
  - a writer knowingly lifts text directly from other authors' work without giving appropriate credit.
- Duplicate publication
  - an author submits for publication a previously published work as if it were original.
- Self-plagiarism
  - a writer copies large parts of an earlier manuscript word for word into a new manuscript.
  - can occur when individuals pursue large programs of research over many years on the same topic, so they are constantly building on their own work and in their own language.

Protecting against plagiarism

1. Always acknowledge the sources of and contributions to your ideas.
2. Enclose in quotation marks any passage of text that is directly taken from another author’s work and acknowledge that author in an in-text citation.
3. Acknowledge every source you use in writing, whether you paraphrase it, summarise it, or quote it directly.
4. When paraphrasing or summarising other authors’ work, reproduce the meaning of the original author’s ideas or facts as closely as possible using your own words and sentence composition.
5. Do not copy sections of your previously published work into a new manuscript without citing the publication and using quotation marks.

Recognition of co-author contributions

- Concerns the **appropriate acknowledgement** (not too much or too little) of collaborators' substantial contributions to a piece of scholarly work.

- An ethical issue that appears frequently in scientific work because collaboration is the norm, not the exception.
  - Working alone means less productivity
  - Working alone means having to do every thing well
  - Collaboration means sharing workload, complementing skills, broadening the domain of interest

- Recognizing co-author contributions appropriately be difficult to deal with because the correct attribution of credit sounds easy but is hard to identify in practice.

- Making co-authorship decisions is important because on the one hand co-authorship confers credit to individuals for their contribution to academic tasks, which can have academic, social, and financial implications; but on the other hand, co-authorship also implies responsibility and accountability for published works.

Four ethical issues relating to co-authorship

1. Coercion authorship

- Occurs when intimidation is used to gain authorship credit, such as when a senior person pressures a more junior person to include the senior person's name on a paper to which he or she has not contributed enough to qualify for authorship.

2. Gift authorship

- Occurs when individuals are given recognition as co-authors without having made substantial contributions, often for reasons like acknowledging friendship, gaining favour, or giving the paper more legitimacy by adding well-known senior researchers to the list of authors.

Four ethical issues relating to co-authorship

3. Mutual support authorship

- Occurs when two or more authors (or author groups) agree to place their names on each other's papers to enhance their perceived productivity. The "authors" can count both publications towards their own list of papers, receive citations for both papers, and so forth.

4. Ghost authorship

- Occurs when papers are written by people who are not included as authors or are not acknowledged. A typical form of ghost authorship involves using or hiring professional scientific writers, perhaps because the researchers feel they cannot write "well" or "scientifically."

Managing co-authorship

- **Golden rule of publishing:** Good papers built on good research.
  - You can contribute in either or both areas.

- Most important involved in the research process warrant co-author recognition when done by someone else but not all of them!
  - Developing an original idea
  - Designing a study
  - Organizing data collection
  - Collecting data
  - Analyzing data
  - Writing and revising a paper
  - Sponsoring/funding the project
  - Managing the project

My decision rules

- **A co-author has…**
  1. made substantial contributions to the conception or design of the research or the acquisition, analysis, or interpretation of data for the research; and
  2. made substantial contributions to drafting the publication or revising it critically for important intellectual content; and
  3. given final approval of the version to be published; and
  4. agreed to be accountable for all aspects of the work, including being accountable for the parts of the work he or she has done, being able to identify which co-authors are responsible for other parts of the work, and having integrity about the contributions of other co-authors.
- **Authorship of a research output should not be claimed when**
  1. participation rests solely in the acquisition of funding or the collection of data.
  2. General supervision of the research group does not justify authorship.
- **These are my criteria; they may or may not be yours**. Rules change by country, institution, and sometimes persons.

- E.g., the DFG (“Deutsche Forschungsgemeinschaft”) demands “participation in” rather then “contribution to” the points above.

Exercise: Recognizing co-authorship

- Professor Smith, the head of the lab, is publishing a paper on the structure of chitin.
- Professor Smith's lab collaborated with a high profile lab group in Sweden that had already engineered and published the correct gene construct to express chitin in vitro, and who sent some of their materials to help Professor Smith's team.
- Professor Smith's post-doc, Mary, did the majority of the lab work, staying late and working long hours to get the necessary data. A final year PhD student, Jiang, and a technician, Oliver, both helped Mary do some of the technical work.
- Professor Smith did not write any of the paper, but reviewed and edited Mary's drafts that she sent to him. He is writing the cover letter and submitting the paper to Nature.
- Mary wrote the bulk of the paper but for the Introduction she used paragraphs of text directly from Jiang's unsubmitted, draft thesis.

Who should be listed as a co-author?

Authorship credit: what is the order of authors?

- The order of authors is entirely up to the authors.
- Typical practices when it comes to putting the authors’ names on the paper…
  - Most often, the first authors are the ones who did most of the work, with the authors listed in descending order of contributions
  - Some put principal investigators at the end of the list
  - Some groups do it alphabetically
  - In some scientific fields the most highly credited author is the one whose name appears [first/last].
  - Social scientists tend to place the authors’ names in alphabetical order regardless of the amount of effort that was contributed.
  - Some journals allow annotations identifying one or two authors who did the majority of the work (this can be important for PhD thesis defenses or for job applications)

A formal way of handling co-auhtor recognition: Credit

- Credit (Contributor Roles Taxonomy)
- A system introduced with the intention of recognizing individual author contributions, reducing authorship disputes and facilitating collaboration.
- Defines a set of roles that individuals can occupy in a research process
- Credit statements can be included in submissions or in final papers
- Example Credit statement
  - Zhang San: Conceptualization, Methodology, Software Priya Singh.: Data curation, Writing- Original draft preparation. Wang Wu: Visualization, Investigation. Jan Jansen: Supervision.: Ajay Kumar: Software, Validation.: Sun Qi: Writing- Reviewing and Editing.

Credit Roles

Term	Definition
Conceptualization	Ideas; formulation or evolution of overarching research goals and aims
Methodology	Development or design of methodology; creation of models
Software	Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code
Validation	Verification, whether as a part of the activity or separate, of the overall replication/ reproducibility of results/experiments and other research outputs
Formal analysis	Application of statistical, mathematical, computational, or other formal techniques to analyze or synthesize study data
Investigation	Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection
Resources	Provision of study materials, reagents, materials, patients, laboratory samples, animals, instrumentation, computing resources, or other analysis tools
Data Curation	Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is necessary for interpreting the data itself) for initial use and later reuse
Writing - Original Draft	Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation)
Writing - Review & Editing	Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre-or postpublication stages
Visualization	Preparation, creation and/or presentation of the published work, specifically visualization/ data presentation
Supervision	Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team
Project administration	Management and coordination responsibility for the research activity planning and execution
Funding acquisition	Acquisition of the financial support for the project leading to this publication

Managing co-authorship: Communicate early and openly

- Open lines of communication throughout the research process are vital.
  - You should talk openly and frankly.
  - It's the team's responsibility to create a communication environment without fear of reprisal, demotion, or other punishment.

- Most important part of this process:
  - Voicing one's investment,
  - Creating transparency about publication strategies,
  - Mutually recognizing each other's goals,
  - Building flexibility into the process, and
  - Establishing commonly accepted criteria for making these basic

Honest reporting

- An ethical standard that demands that research publications comply with expectations for transparency, openness, and reproducibility.
- Typical ethical issues
  - Publication bias
    - systematic suppression of a certain type of research results in published papers, such as negative hypothesis tests or replications, or
    - systematic preference given to innovative and novel findings rather than confirmations of known findings
  - HARKing (hypothesizing after the results are known)
  - *p*-Hacking (the misuse of data analysis to find patterns in data that can be presented as statistically significant)

Publication bias

- occurs when the outcome of an experiment or research study influences the decision whether to publish or otherwise distribute it.
- Known as the file-drawer problem: often investigators decline to submit results when they are found not to support initial hypotheses
- Consequence: the publication of “negative” or “insignificant” results is impeded
- Then, published studies are no longer a representative sample of the available evidence.

Hacking

- The false portrayal of a post hoc hypothesis as if it were an a priori hypothesis.
  - Can invalidates the idea of a priori hypothesis generation and subsequent testing.
  - Can lead to scholars not communicating valuable information about what did not work
  - Can lead to distorted publications limited to ideas and findings without a faithful representation of the scientific process through which these ideas were born.
  - Risks increasing levels of Type 1 errors: if one attempts (too) many post hoc analyses on the same data, some tests will generate false positives simply by chance
  - Risks favoring weaker theories that post hoc accommodate results rather than correctly predict them.

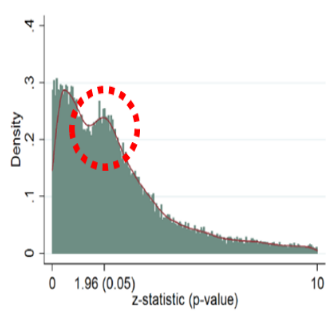
p-Hacking

- occurs when researchers collect or select data or statistical analyses until nonsignificant results become significant.
- Significant results increase the chance of being published but when published data are biased, data synthesis might lead to flawed conclusions.
- Means that we do not know if the strength of the relationship found is purely an artifact of the sample, the analytical method used, or legitimate judgment calls made by the researcher.

Economics

Brodeur et al (AEJ:A, in press)

"Star Wars: The empirics strike back"

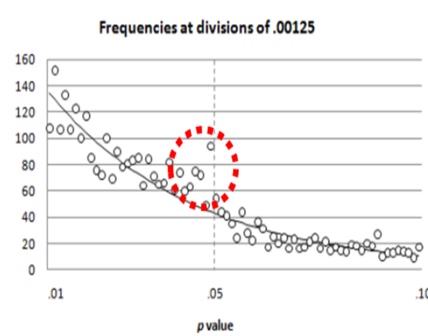


(b) De-rounded distribution of z-statistics.

Psychology

Masicampo Lalande (QJEP, 2012)

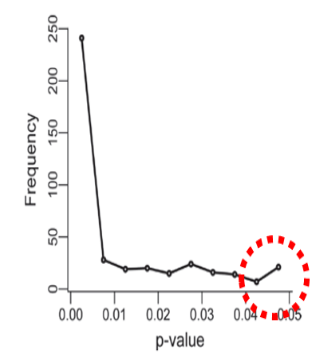
"A peculiar prevalence of p values just below .05"



Biology

Head et al (PLOS Biology 2015)

"Extent and Consequences of P-Hacking in Science"



Recommendations for honest reporting

- **Pre-registration**: make hypotheses public prior to data colletion and analysis
- **Open science**: publicly share raw data used in analysis
- **Register procedures**: Make analyses (e.g., codes, programs) available to others
- **Conduct replications**: repeat studies to see if results remain robust

Appropriate use of language

- Refers to the wording of scientific reports so they are not biased in terms of gender, race, orientation, culture, or any other characteristics.
- Stipulates using gender-responsible, ethnicity-responsible, and inclusive language wherever possible.
- Guidelines:
  - **Specificity**
    - describe specific behaviours rather than stereotypes: e.g., calling a behavior “dominant and opinionated” instead of “typically male”.
  - **Labelling**
    - Refer to concrete labels rather than abstract class tags, e.g., referring to countries’ populations—Mexicans or Chinese—instead of classes like “Hispanics” or “Asians”
  - **Professional acknowledgments**
    - Use professional classifications, not personal labels, like “medical practitioner” or “doctor” instead of “female doctor.”

Responding to Allegations of Possible Misconduct

- Journals should have a clear policy on handling concerns or allegations about misconduct, which can arise regarding authors, reviewers, editors, and others. Journals do not have the resources or authority to conduct a formal judicial inquiry or arrive at a formal conclusion regarding misconduct. That process is the role of the individual's employer, university, granting agency, or regulatory body. However, journals do have a responsibility to help protect the integrity of the public scientific record by sharing reasonable concerns with authorities who can conduct such an investigation.
-
- Deception may be deliberate, by reckless disregard of possible consequences, or by ignorance. Since the underlying goal of misconduct is to deliberately deceive others as to the truth, the journal's preliminary investigation of potential misconduct must take into account not only the particular act or omission, but also the apparent intention (as best it can be determined) of the person involved. Misconduct does not include unintentional error.

Responding to Allegations of Possible Misconduct

- The most common forms of scientific misconduct include (the following are taken with minor modification from the ORI publication Analysis of Institutional Policies for Responding to Allegations of Scientific Misconduct [http://ori.dhhs.gov/html/polanal2.htm, full report in PDF format, accessed 3/13/04]):
- **Falsification of data:** ranges from fabrication to deceptive selective reporting of findings and omission of conflicting data, or willful suppression and/or distortion of data.
- **Plagiarism:** The appropriation of the language, ideas, or thoughts of another without crediting their true source, and representation of them as one's own original work.
- **Improprieties of authorship:** Improper assignment of credit, such as excluding others, misrepresentation of the same material as original in more than one publication, inclusion of individuals as authors who have not made a definite contribution to the work published; or submission of multi-authored publications without the concurrence of all authors.

Responding to Allegations of Possible Misconduct

- **Misappropriation of the ideas of others:** an important aspect of scholarly activity is the exchange of ideas among colleagues. Scholars can acquire novel ideas from others during the process of reviewing grant applications and manuscripts. However, improper use of such information can constitute fraud. Wholesale appropriation of such material constitutes misconduct.
- **Violation of generally accepted research practices:** Serious deviation from accepted practices in proposing or carrying out research, improper manipulation of experiments to obtain biased results, deceptive statistical or analytical manipulations, or improper reporting of results.
- **Material failure to comply with legislative and regulatory requirements affecting research:** Including but not limited to serious or substantial, repeated, willful violations of applicable local regulations and law involving the use of funds, care of animals, human subjects, investigational drugs, recombinant products, new devices, or radioactive, biologic, or chemical materials.
- **Inappropriate behavior in relation to misconduct:** this includes unfounded or knowingly false accusations of misconduct, failure to report known or suspected misconduct, withholding or destruction of information relevant to a claim of misconduct and retaliation against persons involved in the allegation or investigation.
- Deliberate misrepresentation of qualifications, experience, or research accomplishments to advance the research program, to obtain external funding, or for other professional advancement.
-

Responses to possible misconduct

- Journals should have an explicit policy describing the process by which they will respond to allegations of misconduct. The process described in the following 2 paragraphs is an example of a policy for an individual journal:
- *All allegations of misconduct will be referred to the Editor-In-Chief, who will review the circumstances in consultation with the deputy editors. Initial fact-finding will usually include a request to all the involved parties to state their case, and explain the circumstances, in writing. In questions of research misconduct centering on methods or technical issues, the Editor-In-Chief may confidentially consult experts who are blinded to the identity of the individuals, or if the allegation is against an editor, an outside editor expert. The Editor-In-Chief and deputy editors will arrive at a conclusion as to whether there is enough evidence to lead a reasonable person to believe there is a possibility of misconduct. Their goal is not to determine if actual misconduct occurred, or the precise details of that misconduct.*
- *When allegations concern authors, the peer review and publication process for the manuscript in question will be halted while the process above is carried out. The investigation described above will be completed even if the authors withdraw their paper, and the responses below will still be considered. In the case of allegations against reviewers or editors, they will be replaced in the review process while the matter is investigated.*
- All such allegations should be kept confidential; the number of inquiries and those involved should be kept to the minimum necessary to achieve this end. Whenever possible, references to the case in writing should be kept anonymous.

Responses to possible misconduct

- Journals have an obligation to readers and patients to ensure that their published research is both accurate and adheres to the highest ethical standard. Therefore, if the inquiry concludes there is a reasonable possibility of misconduct, responses should be undertaken, chosen in accordance with the apparent magnitude of the misconduct. Responses may be applied separately or combined, and their implementation should depend on the circumstances of the case as well as the responses of the participating parties and institutions.

The following options are ranked in approximate order of severity:

- A letter of explanation (and education) sent only to the person against whom the complaint is made, where there appears to be a genuine and innocent misunderstanding of principles or procedure.
- A letter of reprimand to the same party, warning of the consequences of future such instances, where the misunderstanding appears to be not entirely innocent.
- A formal letter as above, including a written request to the supervising institution that a investigation be carried out and the findings of that inquiry reported in writing to the journal.
- Publication of a notice of redundant or duplicate publication or plagiarism, if appropriate (and unequivocally documented). Such publication will not require approval of authors, and should be reported to their institution.
- Formal withdrawal or retraction of the paper from the scientific literature, published in the journal, informing readers and the indexing authorities (National Library of Medicine, etc), if there is a formal finding of misconduct by an institution. Such publication will not require approval of authors, should be reported to their institution, and should be readily visible and identifiable in the journal. It should also meet other requirements established by the International Committee of Journal Editors (accessed 12/2/03). It is recommended that editors inform readers and authors of their reservation of the right to publish a retraction if it meets these conditions, thereby helping decrease arguments with authors.
- Editors or reviewers who are found to have engaged in scientific misconduct should be removed from further association with the journal, and this fact reported to their institution.

Principles of Transparency and Best Practice in Scholarly Publishing

Introduction

- The Committee on Publication Ethics (COPE), the Directory of Open Access Journals (DOAJ), the Open Access Scholarly Publishing Association (OASPA), and the World Association of Medical Editors (WAME) are scholarly organizations that have collaborated to identify principles of transparency and best practice for scholarly publications.
- Editorial decisions should be based on scholarly merit. They should not be affected by the origins of the manuscript, including the nationality, ethnicity, political beliefs, race, or religion of the authors. Journals should ensure no policies create an exclusionary environment for anyone wanting to engage with the journal and should regularly assess their policies for inclusivity.
- Journal content (Name of journal, Website, Publishing schedule, Archiving, Copyright, Licensing)
- Journal practices (Publication ethics, Peer review, Access)
- Organization (Ownership and management, Advisory body, Editorial team/contact information)
- Business practices (Author fees, Other revenue, Advertising, Direct marketing)
- **Translations** (Português, Bengali, Prevod, Latin American Spanish [version 3.0])

Principles of Transparency and Best Practice in Scholarly Publishing

Journal content

1. Name of journal
2. Website
3. Publishing schedule
4. Archiving
5. Copyright
6. Licensing

Journal practices

7. Publication ethics and related editorial policies
8. Peer review
9. Access

Organization

10. Ownership and management
11. Advisory body
12. Editorial team/contact information

Business practices

13. Author fees
14. Other revenue
15. Advertising
16. Direct marketing